in which m is exactly equal to the applied force $3F_0$ multiplied by the length of the column L and V is exactly equal to the applied force $3F_0$.

Example 4.3 Static Analysis of SDOF Column with Input Displacement (1)
Consider that the SDOF column with the length $L = 3$ m extracted from a RC frame with two RHs and one SH shown in Figure 4.9(a), the sectional parameters are shown in Figure 4.9(b). The SDOF column is subjected by the input lateral displacement x and the axial load $P = 500$ kN. The concrete strength is 30 MPa, and the yield strengths of the longitudinal reinforcement and transverse reinforcement are 335 MPa and 235 MPa, respectively. The symmetrical parameters of the bending and shear primary curves are listed in Table 4.1 and Table 4.2. The input displacement x is specified as 2 mm.

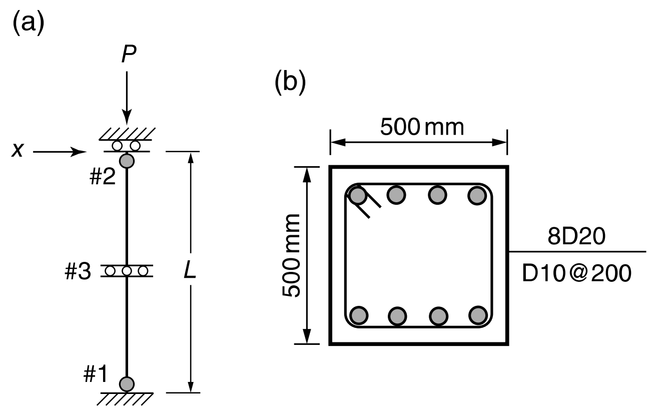

Figure 4.9 The SDOF column with two RHs and one SH: (a) Numerical model; (b) Sectional parameters.

Table 4.1 The parameters of the bending primary curve.

m_{cr} (kN-m)	θ_{cr} (rad)	m_y (kN-m)	θ_y (rad)	θ_u (rad)	θ_f (rad)
91.95	0.00032	229.5	0.0024	0.0165	0.036

Table 4.2 The parameters of the shear primary curve.

V_{cr} (kN)	τ_{cr} (m)	V_y (kN)	τ_y (m)	τ_u (m)	τ_f (m)
61.3	0.00008	153	0.0025	0.0075	0.012

The global stiffness matrices of the SDOF column can be written as:

$$\mathbf{K} = \left[\frac{12}{1+\chi}\frac{EI}{L^3} \right]$$

$$\mathbf{K}' = \left[-\frac{6}{1+\chi}\frac{EI}{L^2} \quad -\frac{6}{1+\chi}\frac{EI}{L^2} \quad \frac{12}{1+\chi}\frac{EI}{L^3} \right]$$

$$\mathbf{K}'' = \begin{bmatrix} \frac{4+\chi}{1+\chi}\frac{EI}{L} & \frac{2-\chi}{1+\chi}\frac{EI}{L} & -\frac{6}{1+\chi}\frac{EI}{L^2} \\ \frac{2-\chi}{1+\chi}\frac{EI}{L} & \frac{4+\chi}{1+\chi}\frac{EI}{L} & -\frac{6}{1+\chi}\frac{EI}{L^2} \\ -\frac{6}{1+\chi}\frac{EI}{L^2} & -\frac{6}{1+\chi}\frac{EI}{L^2} & \frac{12}{1+\chi}\frac{EI}{L^3} \end{bmatrix} \tag{4.34}$$

where $\chi = 12EIb/GAL^2$. The shear effect is introduced to the stiffness matrices $\mathbf{K}$, $\mathbf{K}'$ and $\mathbf{K}''$ in Eq. (4.34) by the parameter χ, and the procedure of obtaining the stiffness matrices can be found as below. By specifying $E = 30$ GPa and $G = 11.5$ GPa, the governing equation can be written as:

$$\left[\begin{array}{c:ccc} 0.6391 & -0.9586 & -0.9586 & 0.6391 \\ \hdashline -0.9586 & 1.9587 & 0.9171 & -0.9586 \\ -0.9586 & 0.9171 & 1.9587 & -0.9586 \\ 0.6391 & -0.9586 & -0.9586 & 0.6391 \end{array}\right] \times 10^5 \left\{\begin{array}{c} x \\ \hdashline -\theta_1'' \\ -\theta_2'' \\ -\tau'' \end{array}\right\} = \left\{\begin{array}{c} F \\ \hdashline m_1 \\ m_2 \\ V \end{array}\right\} \tag{4.35}$$

First, assume that the SDOF column is in the elastic domain, this gives $\theta_1'' = \theta_2'' = 0$ and $\tau'' = 0$. Then the restoring force F, shear force V, moment m_1 and m_2 can be calculated using Eq. (4.35):

$$\begin{Bmatrix} F \\ m_1 \\ m_2 \\ V \end{Bmatrix} = \begin{Bmatrix} 127.8119\,\text{kN} \\ -191.7178\,\text{kN-m} \\ -191.7178\,\text{kN-m} \\ 127.8119\,\text{kN} \end{Bmatrix} \tag{4.36}$$

Comparing these moment and shear demands in Eq. (4.36) with the cracking and yielding capacities in Tables 4.1 and 4.2, the SDOF column is in the plastic domain, which means the assumption that the SDOF column is in the elastic domain is incorrect. Therefore, assume that the structure is in the plastic domain and is located in the hardening branches, which means that the RHs and SH begin to deform in the hardening branches. The $k_f = 2.9\times10^5$ kN m/rad, $\alpha_f = 6.6\times10^4$ kN m/rad, $k_s = 7.7\times10^5$ kN/m and $\alpha_s = 3.8\times10^4$ kN/m can be computed according to the parameters illustrated in Tables 4.1 and 4.2. The moments m_1 and m_2 and shear force V can be expressed in the terms of θ_1'', θ_2'' and τ'' as:

$$\begin{cases} m_1 = -91.95 + 8.6 \times 10^4\, \theta_1'' \\ m_2 = -91.95 + 8.6 \times 10^4\, \theta_2'' \\ V = 61.3 + 4.0 \times 10^4\, \tau'' \end{cases} \tag{4.37}$$

where the minus sign in Eq. (4.37) denotes that the negative cracking capacities has been exceeded. Therefore, Eq. (4.35) becomes:

$$\left[\begin{array}{c|ccc} 0.6391 & -0.9586 & -0.9586 & 0.6391 \\ \hline -0.9586 & 1.9587+0.86 & 0.9171 & -0.9586 \\ -0.9586 & 0.9171 & 1.9587+0.86 & -0.9586 \\ 0.6391 & -0.9586 & -0.9586 & 0.6391+0.40 \end{array}\right] \times 10^5 \left\{\begin{array}{c} 0.002 \\ \hline -\theta_1'' \\ -\theta_2'' \\ -\tau'' \end{array}\right\} = \left\{\begin{array}{c} F \\ \hline -91.95 \\ -91.95 \\ 61.3 \end{array}\right\} \tag{4.38}$$

Solving for the plastic deformations θ_1'', θ_2'' and τ'' in Eq. (4.38) gives:

$$\begin{Bmatrix} \theta_1'' \\ \theta_2'' \\ \tau'' \end{Bmatrix} = \begin{Bmatrix} -0.1952\,\text{rad} \\ -0.1952\,\text{rad} \\ 0.2804\,\text{m} \end{Bmatrix} \times 10^{-3} \tag{4.39}$$

Substituting Eq. (4.39) back into Eq. (4.35) and solving for the restoring force F, the moments m_1 and m_2 at the RHs and the shear force V gives:

$$\begin{Bmatrix} F \\ m_1 \\ m_2 \\ V \end{Bmatrix} = \begin{Bmatrix} 72.4767\,\text{kN} \\ -108.7150\,\text{kN-m} \\ -108.7150\,\text{kN-m} \\ 72.4767\,\text{kN} \end{Bmatrix} \tag{4.40}$$

Comparing these demands in Eq. (4.40) with the corresponding capacities in Tables 4.1 and 4.2, the structure is in the plastic domain and the RHs and SH behave on the hardening branches, which means that the latest assumption is made correctly and the solution in Eq. (4.40) are the final results.

Example 4.4 Static Analysis of SDOF Column with Input Displacement (2)
Consider again the SDOF column as shown in Figure 4.9 with the same parameters in Tables 4.1 and 4.2. Now let the input displacement be equal to 20 mm.

According to results obtained in Example 4.3, first assume that the SDOF column is in the plastic domain and the two RHs and one SH behave in the hardening branches. Then Eq. (4.38) can be rewritten here by specifying the total displacement x as 0.02:

$$\left[\begin{array}{c:ccc} 0.6391 & -0.9586 & -0.9586 & 0.6391 \\ \hdashline -0.9586 & 1.9587+0.86 & 0.9171 & -0.9586 \\ -0.9586 & 0.9171 & 1.9587+0.86 & -0.9586 \\ 0.6391 & -0.9586 & -0.9586 & 0.6391+0.40 \end{array}\right] \times 10^5 \left\{\begin{array}{c} 0.02 \\ \hdashline -\theta_1'' \\ -\theta_2'' \\ -\tau'' \end{array}\right\} = \left\{\begin{array}{c} F \\ \hdashline -91.95 \\ -91.95 \\ 61.3 \end{array}\right\} \tag{4.41}$$

Solving for the plastic deformations θ_1'', θ_2'' and τ'' in Eq. (4.41) gives:

$$\begin{Bmatrix} \theta_1'' \\ \theta_2'' \\ \tau'' \end{Bmatrix} = \begin{Bmatrix} -0.0036\,\text{rad} \\ -0.0036\,\text{rad} \\ 0.0051\,\text{m} \end{Bmatrix} \tag{4.42}$$

Substituting Eq. (4.42) back into Eq. (4.35) and solving for the restoring force F, the moments m_1 and m_2 at the RHs and the shear force V gives:

$$\begin{Bmatrix} F \\ m_1 \\ m_2 \\ V \end{Bmatrix} = \begin{Bmatrix} 265.7743\,\text{kN} \\ -398.6614\,\text{kN-m} \\ -398.6614\,\text{kN-m} \\ 265.7743\,\text{kN} \end{Bmatrix} \tag{4.43}$$

The rotations θ_1 and θ_2 and the shear deformation τ of the structure as illustrated in Figure 4.7 can be calculated according to the displacement decomposition expressed in Section 4.1.2:

$$\theta_1 = \theta_2 = -0.0050\,\text{rad}, \quad \tau = 0.0054\,\text{m} \tag{4.44}$$

Comparing the demands in Eqs. (4.43) and (4.44) with the corresponding capacities in Tables 4.1 and 4.2, the structure should be represent by the yielding platform.

The α_f and α_s here are specified by 0.1 that close to zero to simulate the yielding platform branches. Then the moments m_1 and m_2 and shear force V can be expressed in the terms of θ_1'', θ_2'' and τ'' according to Eq. (4.5):

$$\begin{cases} m_1 = -229.4998 + 0.1\,\theta_1'' \\ m_2 = -229.4998 + 0.1\,\theta_2'' \\ V = 152.9998 + 0.1\,\tau'' \end{cases} \tag{4.45}$$

After adding the tangent stiffnesses expressed in Eq. (4.45) to the stiffness matrix $\mathbf{K}''$, Eq. (4.41) becomes:

$$\left[\begin{array}{c:ccc} 0.6391 & -0.9586 & -0.9586 & 0.6391 \\ \hdashline -0.9586 & 1.958701 & 0.9171 & -0.9586 \\ -0.9586 & 0.9171 & 1.958701 & -0.9586 \\ 0.6391 & -0.9586 & -0.9586 & 0.639101 \end{array}\right] \times 10^5 \left\{\begin{array}{c} 0.02 \\ \hdashline -\theta_1'' \\ -\theta_2'' \\ -\tau'' \end{array}\right\} = \left\{\begin{array}{c} F \\ \hdashline -229.4998 \\ -229.4998 \\ 152.9998 \end{array}\right\} \tag{4.46}$$

Solving for the plastic deformations θ_1'', θ_2'' and τ'' in Eq. (4.46) gives:

$$\begin{Bmatrix} \theta_1'' \\ \theta_2'' \\ \tau'' \end{Bmatrix} = \begin{Bmatrix} -0.0045\,\text{rad} \\ -0.0045\,\text{rad} \\ 0.0042\,\text{m} \end{Bmatrix} \tag{4.47}$$

Substituting Eq. (4.47) back into Eq. (4.35) and solving for the restoring force F, the bending moments m_1 and m_2 at the RHs and the shear force V gives:

$$\begin{Bmatrix} F \\ m_1 \\ m_2 \\ V \end{Bmatrix} = \begin{Bmatrix} 153.0002\,\text{kN} \\ -229.5003\,\text{kN-m} \\ -229.5003\,\text{kN-m} \\ 153.0002\,\text{kN} \end{Bmatrix} \tag{4.48}$$

in which the results are quite close to the yielding capacities m_y and V_y listed in Tables 4.1 and 4.2, which means that the yielding platform behaviors can be well simulated. The rotations θ_1 and θ_2 and the shear deformation τ of the structure can be obtained as:

$$\theta_1 = \theta_2 = -0.0053\ \text{rad}, \quad \tau = 0.0044\ \text{m} \tag{4.49}$$

Comparing the demands in Eq. (4.48) and Eq. (4.49) with the corresponding capacities in Tables 4.1 and 4.2, the structure is in the plastic domain and the RHs and SH behave on the yielding platforms, which means that the latest assumption is made correctly and the solution in Eq. (4.48) are the final results.

Example 4.5 Static Analysis of SDOF Column with Input Displacement (3)

Consider again the SDOF column as shown in Figure 4.9 with the same parameters in Tables 4.1 and 4.2. Now let the input displacement be equal to 60 mm.

According to the results obtained in Example 4.4, first assume that the SDOF column is in the plastic domain and the two RHs and one SH behave in the yielding platform branches. Then Eq. (4.46) can be rewritten here by specifying the total displacement x as 0.06:

$$\left[\begin{array}{c:ccc} 0.6391 & -0.9586 & -0.9586 & 0.6391 \\ \hdashline -0.9586 & 1.958701 & 0.9171 & -0.9586 \\ -0.9586 & 0.9171 & 1.958701 & -0.9586 \\ 0.6391 & -0.9586 & -0.9586 & 0.639101 \end{array}\right] \times 10^5 \left\{\begin{array}{c} 0.06 \\ \hdashline -\theta_1'' \\ -\theta_2'' \\ -\tau'' \end{array}\right\} = \left\{\begin{array}{c} F \\ \hdashline -229.4998 \\ -229.4998 \\ 152.9998 \end{array}\right\} \tag{4.50}$$

Solving for the plastic deformations θ_1'', θ_2'' and τ'' in Eq. (4.50) gives:

$$\begin{Bmatrix} \theta_1'' \\ \theta_2'' \\ \tau'' \end{Bmatrix} = \begin{Bmatrix} -0.0154\,\text{rad} \\ -0.0154\,\text{rad} \\ 0.0115\,\text{m} \end{Bmatrix} \tag{4.51}$$

Substituting Eq. (4.51) back into Eq. (4.50) and solving for the restoring force F, the moments m_1 and m_2 at the RHs and the shear force V gives:

$$\begin{Bmatrix} F \\ m_1 \\ m_2 \\ V \end{Bmatrix} = \begin{Bmatrix} 153.0009\text{ kN} \\ -229.5014\text{ kN-m} \\ -229.5014\text{ kN-m} \\ 153.0009\text{ kN} \end{Bmatrix} \tag{4.52}$$

Then, the rotations θ_1 and θ_2 at the column ends and the shear deformation τ of the structure can be obtained:

$$\theta_1 = \theta_2 = -0.0162\text{ rad}, \quad \tau = 0.0117\text{ m} \tag{4.53}$$

Comparing the demands in Eqs. (4.52) and (4.53) with the corresponding capacities in Tables 4.1 and 4.2, we can find that the strength of SH starts to deteriorate. However, due to the same primary curve parameters of the two RHs shown in Table 4.1 and the assumption of the procedure of dividing the total primary curve to the bending and shear ones for the SDOF system in previous section, the RHs and SH should behave on the same type of branches at the same force level. Therefore, assume that the RHs and the SH behave on the corresponding softening branches and the moments m_1 and m_2 and shear force V can be expressed in the terms of θ_1'', θ_2'' and τ'' according to Eq. (4.5):

$$\begin{cases} m_1 = -407.02 - 1.1\times 10^4\,\theta_1'' \\ m_2 = -407.02 - 1.1\times 10^4\,\theta'' \\ V = 390.67 - 3.3\times 10^4\,\tau'' \end{cases} \tag{4.54}$$

Therefore, Eq. (4.50) becomes:

$$\left[\begin{array}{c|ccc} 0.6391 & -0.9586 & -0.9586 & 0.6391 \\ \hline -0.9586 & 1.9587-0.11 & 0.9171 & -0.9586 \\ -0.9586 & 0.9171 & 1.9587-0.11 & -0.9586 \\ 0.6391 & -0.9586 & -0.9586 & 0.6391-0.33 \end{array}\right] \times 10^5 \left\{\begin{array}{c} 0.06 \\ \hline -\theta_1'' \\ -\theta_2'' \\ -\tau'' \end{array}\right\} = \left\{\begin{array}{c} F \\ \hline -407.02 \\ -407.02 \\ 390.67 \end{array}\right\} \tag{4.55}$$

Solving for the plastic deformations θ_1'', θ_2'' and τ'' in Eq. (4.55) gives:

$$\begin{Bmatrix} \theta_1'' \\ \theta_2'' \\ \tau'' \end{Bmatrix} = \begin{Bmatrix} -0.0167\text{ rad} \\ -0.0167\text{ rad} \\ 0.0075\text{ m} \end{Bmatrix} \tag{4.56}$$

Substituting Eq. (4.56) back into Eq. (4.35) and solving for the restoring force F, the moments m_1 and m_2 at the RHs and the shear force V gives:

$$\begin{Bmatrix} F \\ m_1 \\ m_2 \\ V \end{Bmatrix} = \begin{Bmatrix} 145.25\text{ kN} \\ -217.88\text{ kN-m} \\ -217.88\text{ kN-m} \\ 145.25\text{ kN} \end{Bmatrix} \tag{4.57}$$

Then, the rotation deformations θ_1 and θ_2 at the column ends and the shear deformation τ of the structure can be obtained:

$$\theta_1 = \theta_2 = -0.0175\,\text{rad}, \quad \tau = 0.0077\,\text{m} \tag{4.58}$$

Comparing the demands in Eqs. (4.57) and (4.58) with the corresponding capacities in Tables 4.1 and 4.2, the structure is in the plastic domain and the RHs and SH behave on the softening branches, which means that the latest assumption is made correctly and the solution in Eq. (4.57) are the final results.

4.3.2 Force Analogy Method for Static Multi-Degree-of-Freedom Systems

For a multi-degree-of-freedom (MDOF) system with n degrees of freedom (DOFs), the displacement can be written in vector form as:

$$\mathbf{x} = \mathbf{x}' + \mathbf{x}'' = \begin{Bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{Bmatrix} + \begin{Bmatrix} x''_1 \\ x''_2 \\ \vdots \\ x''_n \end{Bmatrix} \tag{4.59}$$

where $\mathbf{x}$ represents the total displacement vector, $\mathbf{x}'$ is the elastic displacement vector, and $\mathbf{x}''$ is the inelastic displacement vector.

Consider a RC frame constructed by flexural members, the q_f RHs are located at two ends of the members to represent the plastic bending behavior and q_s SHs are assigned to the columns to represent the plastic shear behavior. Then the moment vector $\mathbf{m}$ at the RHs and shear force vector $\mathbf{V}$ according to the SHs can be described by the force vector $\mathbf{f}$ as:

$$\mathbf{f} = \begin{Bmatrix} \mathbf{m} \\ \hdashline \mathbf{V} \end{Bmatrix} = \begin{Bmatrix} \mathbf{m}' \\ \hdashline \mathbf{V}' \end{Bmatrix} + \begin{Bmatrix} \mathbf{m}'' \\ \hdashline \mathbf{V}'' \end{Bmatrix} = \begin{Bmatrix} m'_1 \\ \vdots \\ m'_{q_f} \\ \hdashline V'_{q_f+1} \\ \vdots \\ V'_{q_f+q_s} \end{Bmatrix} + \begin{Bmatrix} m''_1 \\ \vdots \\ m''_{q_f} \\ \hdashline V''_{q_f+1} \\ \vdots \\ V''_{q_f+q_s} \end{Bmatrix} \tag{4.60}$$

where $\mathbf{m}'$ and $\mathbf{V}'$ are the vectors of elastic moments and elastic shear forces respectively due to elastic displacement $\mathbf{x}'$, and $\mathbf{m}''$ and $\mathbf{V}''$ are the vectors of inelastic moments and shear forces respectively due to inelastic displacement $\mathbf{x}''$.

Corresponding to the inelastic moment and shear force vectors $\mathbf{m}''$ and $\mathbf{V}''$, the inelastic deformations θ'' and τ'' can be written in vector form as: